

Zoë Fisk

(860) 333-8831 | zoejewelfisk@gmail.com | github.com/zoefisk | linkedin.com/in/zjfish | zoefisk.com | Seattle, WA

SUMMARY

Full-stack developer who builds offline-first platforms, custom data visualization tools, and web applications that transform complex information into intuitive software. Develops functional prototypes from the ground up, combining web engineering, Agile project leadership, and hands-on technical development. Creates tools that streamline workflows, reduce data-processing time, and simplify complex user experiences.

SKILLS

Programming Languages : Java, C#, C++, Python, TypeScript, JavaScript, HTML/CSS, Lua, R

Frameworks & Libraries : React, React Native, Node.js, Next.js, Express.js, D3.js, Vega, Vega-Lite, Material UI, Mantine, Tailwind CSS, Zod, D3.js, Vega, Vega-Lite, Three.js

Databases : MongoDB, Firebase, PostgreSQL, Oracle

Tools & Platforms : Git, GitHub, Vercel, Auth0, NextAuth, Jira, Figma, VS Code, JetBrains IDEs, Notion, Trello

AI Tools : ChatGPT, Claude Code, Gemini, Prompt Engineering, AI-Assisted Development, LLM Workflow Design

Concepts : Agile Development, Scrum, UI/UX Design, User Study Implementation

PROJECTS

FRC Scouting Application (Solo Project) | zoefisk.com/projects/scouting-app Mar. 2026 – Present

- Replaced manual competitive intelligence profiling by architecting an offline-first platform projected to reduce data collection and analysis time by 50% for approximately 20 team members across 60+ qualification matches per competition.
- Generated actionable competitive intelligence through automated scouting schedules, configurable forms, team performance profiles, match histories, rankings, alliance pick lists, and analytics while ensuring reliable offline data collection through synchronization, local caching, and QR code-based peer-to-peer transfer.
- Designed the platform for multi-team adoption by building configurable workflows, collaborating with stakeholders, and integrating an AI chatbot into the custom form builder to streamline scouting form creation.
- Tech Stack:** Next.js, React, TypeScript, Firebase, NextAuth, Vercel, ChatGPT Codex, Zod, The Blue Alliance API, Material UI, AI integration, offline synchronization, and Git/GitHub.

Reference Range Number Line Visualization Framework (MQP) | rrnl-showcase.com Aug. 2025 – May 2026

- Standardized the creation of Reference Range Number Line (RRNL) health visualizations as part of a 4-person team by co-developing a domain-specific language (DSL) and visualization pipeline that transformed structured JSON into consistent, customizable Vega and D3.js visualizations by encoding key design characteristics from existing RRNL literature.
- Developed an interactive web application featuring a live JSON editor, graph customization, real-time previews, and project information.
- Implemented and managed a replicated user study with 123 participants by integrating DSL-generated visualizations into ReVISit, configuring study materials, recruiting participants through Prolific, and overseeing study execution.
- Tech Stack:** Next.js, React, TypeScript, Python, D3.js, Vega, Material UI, JSON, ReVISit, Prolific, Figma, and Git/GitHub.

Hospital Navigation Web Application (9-Person Team) Mar. 2023 – May 2023

- Simplified hospital wayfinding by co-leading frontend development of a hospital navigation prototype for Mass General Brigham Hospital, featuring interactive routes and step-by-step directions for patients and visitors.
- Served as a co-project manager and co-Scrum Master, leading daily stand-ups and coordinating Agile development.
- Tech Stack:** React, TypeScript, Node, Express, Material UI, Tailwind CSS, Figma, Jira, PostgreSQL, Git/GitHub, Webstorm, and AWS.

Interactive Qualifying Project (IQP) | <https://wp.wpi.edu/nantucket/projects/2024-projects/carat/> Aug. 2024 – Dec. 2024

- Positioned Sustainable Nantucket to launch the Community & Agriculture Resilience Audit Tool (CARAT) as part of a 4-person team by conducting research with 21 stakeholders and a focus group across agriculture, education, nonprofits, and government.
- Defined data requirements and stakeholder workflows by benchmarking comparable communities, delivering adapted recommendations and an implementation roadmap through a comprehensive report and televised public presentation.

Item Request Indicator (FoM Video Game Mod) | <https://www.nexusmods.com/fieldsofmistria/mods/1117> Aug. 2026

- Improved quest-item tracking by building and publishing a GML UI mod that surfaces required items and request details within inventory and storage screens, reaching 750+ unique downloads within its first week.

EXPERIENCE

Mosaic Software (WPI Organization) Mar. 2025 – Dec. 2025
Software Engineer (Volunteer) Worcester, MA

- Supported frontend development in React for Terrazzo, an open-source project management and collaboration platform, as part of a small Agile development team.
- Contributed to sprint planning, design discussions, code reviews, and iterative development using React, Git, Trello, and Scrum.

LEADERSHIP & ACTIVITIES

FIRST Robotics Competition (Team #3461)

2016 – 2022

Student Member, Co-Captain

Bristol, CT

- Co-led a 5-student team to win the FIRST Championship Quality Award, earning international recognition among ~3,000 teams for engineering quality and execution.
- Programmed autonomous and driver-controlled robot logic in Java, designed CAD assemblies in SolidWorks and Onshape, and machined custom components.
- Secured team funding through grant proposals and fundraising campaigns, authored FIRST Impact Award submissions, and delivered technical presentations to competition judges.

WPI Computer & Technology Club

2023 – 2026

President (2024 – 2025), Secretary (2023 – 2024, 2025 – 2026)

Worcester, MA

- Improved club operations by organizing shared documentation, standardizing meeting minutes, planning events further in advance, and delegating leadership responsibilities across the officer team.

EDUCATION

Worcester Polytechnic Institute

Aug. 2022 – May 2026

Bachelor of Science in Computer Science with Distinction

Worcester, MA